University of Barishal

Department of Computer Science and Engineering

Course Title: Discrete Mathematics

Course Code: CSE-1203

1st Year 2nd Semester Final Examination

Admission Session: 2021-2022

Marks: 60

Time: 03 Hours

N.B.: Answer any **Five** questions out of the followings. All parts of each question must be answered consecutively. Right side of the question shows the maximum marks.

1.a. Define proposition with example. Consider the following sentences and determine which of the following is a proposition or not. **1 + 4**

(i) Dhaka is the capital of Bangladesh.

(ii) $2 + 3 = 5$

(iii) $x + 2 = 11$

$x + y = y + x$, for every pair of real numbers x and y.

b. Construct a truth table for the following compound propositions, $(p \vee q) \rightarrow (p \wedge q)$, where p and q represent arbitrary propositions. Determine whether this compound proposition is tautology, contradiction, or contingency. Justify your answer. **03**

c. Translate each of these statements into logical expressions using predicates, quantifiers, and logical connectives. Let the domain consist of all people. **04**

(i) Someone in your class can speak Japanese.

(ii) Everyone in your class is friendly.

(iii) There is a person in your class who was not born in Jashore.

(iv) No student in your class has taken a course in logic programming.

2.a) Define predicates and quantifiers with examples. **3**

b) What is free variable? Let Q(x) denotes the statement "x=x+1". What is the truth value of the qualification $\exists x Q(x)$, where domain consists of all real numbers? **4**

c) Using Mathematical induction method show that **5**

$$\sum_{i=0}^{n-1} 2^i(n-i) = 2^{n+1} - n - 2$$

3.a) Show that if A, B and C are sets, then $\overline{A \cap B \cap C} = \overline{A} \cup \overline{B} \cup \overline{C}$. **4**

b) Show that $\forall x\,(P(x) \wedge Q(x))$ and $\forall x\, P(x) \wedge \forall x\, Q(x)$ are logically equivalent. **4**

c) Let A = {0, 1, 2, 3} and define relations R, S and T on A as follows: **4**

R = {(0,0), (0,1), (0,3), (1,0), (1,1), (2,2), (3,0), (3,3)}, S = {(0,0), (0,2), (0,3), (2,3)},

T = {(0, 1), (2, 3)}.

i) Is R reflexive? Symmetric? Transitive?

ii) Is S reflexive? Symmetric? Transitive?

4. a) Use rules of inference to show that the hypotheses **"If it does not rain or if it is not foggy, then the sailing race will be held and the lifesaving demonstration will go on," "If the sailing race is held, then the trophy will be awarded,"** and **"The trophy was not awarded"** imply the conclusion **"It rained."** **04**

b) Prove that, if n is an integer and 3n + 2 is odd, then n is odd? **04**

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c) Define Argument. Express the statements "**Some student in this class has visited Mexico**" and "**Every student in this class has visited either Canada or Mexico**" using predicates and quantifiers. 1+3

5.a) Discuss generalized pigeonhole principle and proof this theorem using proof by contradiction. 3

b) Suppose that there are eight runners in a race. The winner receives a gold medal, the second-place finisher receives a silver medal, and the third-place finisher receives a bronze medal. How many different ways are there to award to these medals, if all possible outcomes of the race can occur and there no ties? 3

c) How many bit strings of length 8 contain at least two 1s and at most two 0s? 3

d) Each user on a computer system has a password, which is five to eight characters long, where each character is an uppercase letter or lowercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there? 3

6.a) *"A patient goes to see a doctor. The doctor performs a test with 99 percent reliability--that is, 99 percent of people who are sick test positive and 99 percent of the healthy people test negative. The doctor knows that only 1 percent of the people in the country are sick. Now the question is:* ***if the patient tests positive, what are the chances the patient is sick?"*** 5

Hints (Bayes's theorem):	$P(A\|B) = \frac{P(B\|A)P(A)}{P(B)}$
P(A) is the probability of event A P(B) is the probability of event B	P(A|B) is the probability of observing event A if B is true P(B|A) is the probability of observing event B if A is true.

Table I: Illustrates the scenario in a hypothetic population of 10,000 people

Test Status	Diseased	Not Diseased	Population
Test +	99	99	198
Test -	1	9801	9802
Total:	100	9900	10,000

b) Let E1 and E2 be events in the sample space S. Then proof the following:

$$P(E1 \cup E2) = P(E1) + P(E2) - P(E1 \cap E2)$$

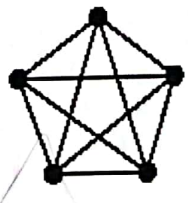
c) What is the probability that a positive integer selected at random from the set of positive integers not exceeding 100 is divisible by either 2 or 5? 3

7.a) A computer company receives 350 applications from college graduates for a job planning a line of new web servers. Suppose that 220 of these applicants majored in CSE, 147 majored in Math, and 51 majored both in CSE and Math. How many of these applicants majored neither in CSE nor in Math. 4

b) Explain Eulerian and Hamiltonian graphs with examples, also draw the graphs of the following: 4

i) Eulerian but not Hamiltonian ii) Hamiltonian but not Eulerian

c) What is planner graph? Draw the planner graph of the given simple graph. 4



8. a) Find all the spanning tree of graph G and find which is the minimal spanning tree of G shown in fig: 06

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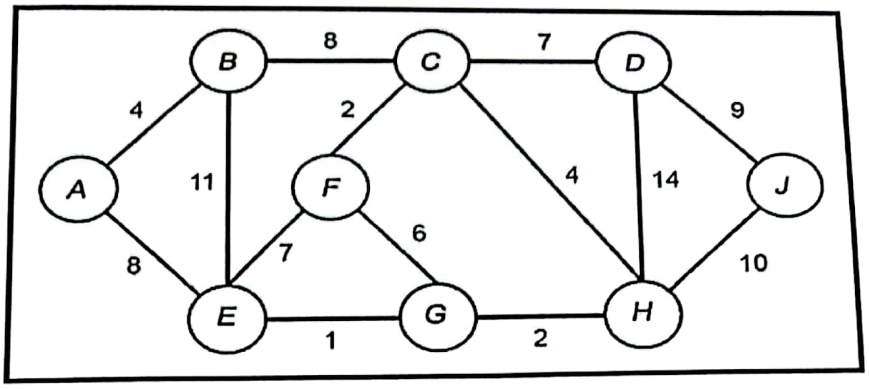

b) Define Directed Graph. A drawing of the directed graph with vertices a, b, c, d, and e, and edges (a, a), (a, b), (b, c), (b, d), (a, d), (a, e), (e, d), (e, e), (e, a), (d, d), (c, e), (c, a) and (b, e). **03**

c) How many 5-digit numbers can be formed by using the digits 0, 1, 2, 3, 4, 5, 6, 7, 8 if every number is to start with '40' with no digit repeated? **03**

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